Lucida

Quick installation on a ready Linux server

Eight steps from a server whose operating system is already installed to examiners using Lucida in Chrome. Package: **Lucida-0.3.0-linux-x86_64.tar.gz**. The full guide (*linux-server-install.pdf*, in the same folder) has more detail on every step, offline driver installation, everyday operation and troubleshooting.

This guide assumes	
Operating system	Ubuntu 22.04 / 24.04 LTS, or Red Hat Enterprise Linux 8 / 9 (also Rocky, Alma), installed, on the network, with a fixed IP address
Hardware	x86_64, 64 GB RAM, NVIDIA L40S
Your account	A user that can use sudo, logged in at the console or with SSH
Network	The server can reach its package repositories (only needed if the NVIDIA driver is not installed yet); examiner PCs can reach the server on TCP 8443
Examiner PCs	Google Chrome; administrator rights once, to trust the server's certificate

Grey boxes are commands: type them on the server unless the box title says otherwise. Green boxes say what you should see. Where Ubuntu and Red Hat differ, the box title says which; follow only yours.

Step 1. Confirm the operating system

```
grep -E '^(NAME|VERSION_ID)=' /etc/os-release
uname -m
hostname -I
```

You should see. *Ubuntu* 22.04 / 24.04, or *Red Hat Enterprise Linux* (or Rocky / AlmaLinux) 8.x / 9.x; `x86_64`; and the server's IP address. Write the address down: examiners will type it in Chrome.

Step 2. Make sure the NVIDIA driver works

```
nvidia-smi
```

You should see. A table naming *NVIDIA L40S* (about 46068 MiB) with *Driver Version* 525 or higher (550+ recommended). If so, skip to step 3. Lucida needs only the driver: do not install the CUDA toolkit or cuDNN.

If `nvidia-smi` is not found, install the driver, then reboot:

Ubuntu

```
sudo apt update
sudo apt install -y ubuntu-drivers-common pciutils
sudo ubuntu-drivers list --gpgpu          # pick the highest ...-server branch it lists
sudo apt install -y nvidia-driver-570-server nvidia-utils-570-server
sudo reboot
```

Red Hat

```
# for version 8, replace every 9 with 8
# (Rocky / Alma 9: sudo dnf config-manager --set-enabled crb instead of the next line)
sudo subscription-manager repos --enable codeready-builder-for-rhel-9-x86_64-rpms
sudo dnf install -y \
  https://dl.fedoraproject.org/pub/epel/epel-release-latest-9.noarch.rpm
sudo dnf install -y kernel-devel-$(uname -r) kernel-headers-$(uname -r) pciutils
sudo dnf config-manager --add-repo \
  https://developer.download.nvidia.com/compute/cuda/repos/rhel9/x86_64/cuda-rhel9.repo
sudo dnf install -y nvidia-open
#   version 8 instead:  sudo dnf module install -y nvidia-driver:open-dkms
sudo reboot
```

Important. If Secure Boot is on (`mokutil --sb-state`), a blue *MOK management* screen appears on the console at the reboot: **Enroll MOK** > **Continue** > **Yes** > the password you chose > **Reboot**. On Red Hat, first run `sudo mokutil --import /var/lib/dkms/mok.pub` to choose that password.

After the reboot, run `nvidia-smi` again, then turn on persistence mode:

```
sudo systemctl enable --now nvidia-persistenced
```

Step 3. Copy the package to the server and check it

Download **Lucida-Linux-Server** from the repository's **Actions** tab (workflow *Linux server package*, newest green run on *main*, *Artifacts*) or from a **Release**, and unzip it on your PC. Then copy both files to the server:

On your PC (PowerShell or Terminal)

```
scp Lucida-0.3.0-linux-x86_64.tar.gz* <your user>@<server address>:~/
```

On the server (if the release came in parts, first: `cat Lucida-0.3.0-linux-x86_64.tar.gz.part* > Lucida-0.3.0-linux-x86_64.tar.gz`):

```
cd ~
sha256sum -c Lucida-0.3.0-linux-x86_64.tar.gz.sha256
tar -xzf Lucida-0.3.0-linux-x86_64.tar.gz
cd Lucida-0.3.0-linux-x86_64
```

You should see. `Lucida-0.3.0-linux-x86_64.tar.gz: OK`. If it says *FAILED*, download it again.

Step 4. Check the server

```
./check-server.sh
```

You should see. Every line **OK** and *All checks passed*. **WARN** lines are allowed (Lucida works); fix any **FAIL** first. The usual ones: an old or missing driver (step 2), or port 8443 in use (then install with `--port 9443` in step 5).

Step 5. Install

```
sudo ./install.sh
```

About five minutes. It installs Lucida in `/opt/lucida` (settings in `/etc/lucida/workbench.ini`, case data in `/var/lib/lucida`), starts the **lucida** service on HTTPS port 8443 with sign-in, opens the port in the firewall, runs the self-test and checks the GPU. It ends like this:

End of the output

```
RESULT: models run on the GPU

== Done
  Open Lucida in Chrome on an examiner PC at:
      https://10.20.30.40:8443
  First install the certificate authority on each examiner PC (once):
      /var/lib/lucida/tls/workbench-ca.crt  (copy: /home/admin/lucida-ca.crt)
      SHA-256 3A:7F:...:C2
  The first person to open Lucida creates the administrator account.
```

To keep the case data on a separate disk, mount it first and add `--data /srv/lucida`. `sudo ./install.sh --help` lists every option.

Step 6. Trust the server's certificate on each examiner PC

Once per PC. Copy the certificate the installer left in your home folder on the server, and compare its SHA-256 with the server's (they must match):

On a Windows PC (PowerShell)

```
scp <your user>@<server address>:lucida-ca.crt .
certutil -hashfile lucida-ca.crt SHA256
```

On the server, to compare

```
sha256sum ~/lucida-ca.crt
```

Then install it, in a PowerShell opened with **Run as administrator**, and restart Chrome:

On a Windows PC, as administrator

```
certutil -addstore -f Root lucida-ca.crt
```

Mac: `sudo security add-trusted-cert -d -r trustRoot -k /Library/Keychains/System.keychain lucida-ca.crt`. Linux PC and Group Policy: see the full guide, step 7.

Step 7. Open Lucida in Chrome and create the administrator

In Chrome's address bar

```
https://<server address>:8443
```

> **You should see.** Lucida's first-start screen, with no certificate warning. If Chrome says *Your connection is not private*, the certificate is not trusted on this PC yet (step 6; restart Chrome completely).

- Create the **administrator** (user name, full name, a password of at least 12 characters). Do it now, before you share the address.
- Your name (top right) > **Users...**: add each examiner with a role (admin, examiner, reviewer, viewer), and a **second administrator** for safe keeping.
- Bookmark the address on each PC.

Step 8. Confirm and record

```
systemctl status lucida      # active (running)
lucida gpu                   # RESULT: models run on the GPU
lucida selftest              # All checks passed.
```

The self-test log is saved in `/var/lib/lucida/logs`; keep it with your lab's validation records, with the package's SHA-256.

If something goes wrong

Problem	Fix
Chrome: connection not private	Step 6 on that PC, then quit and reopen Chrome.
Chrome: connection timed out	Firewall. Ubuntu: `sudo ufw allow 8443/tcp`. Red Hat: `sudo firewall-cmd --permanent --add-port=8443/tcp; sudo firewall-cmd --reload`. Also any network firewall between the PCs and the server.
Chrome: wrong name in certificate, or *Invalid host header*	Add the name or address to `allowed_hosts` in /etc/lucida/workbench.ini, then `sudo systemctl restart lucida`.
Service not running	`journalctl -u lucida -n 80` says why.
`lucida gpu` says CPU	`nvidia-smi` must work (step 2); then `sudo nvidia-modprobe -u -c=0; sudo systemctl restart lucida`.

Sign-off

- ☐ `nvidia-smi` shows the L40S, driver 525 or newer
- ☐ Package SHA-256 checked (step 3) and `check-server.sh` passed
- ☐ `systemctl status lucida`: active (running)
- ☐ `lucida gpu`: models run on the GPU
- ☐ Self-test passed; log kept
- ☐ Certificate trusted on each examiner PC; Chrome opens Lucida with no warning
- ☐ Administrator and a second administrator created; examiners added

Installed by: _____ Date: _____ Server: _____